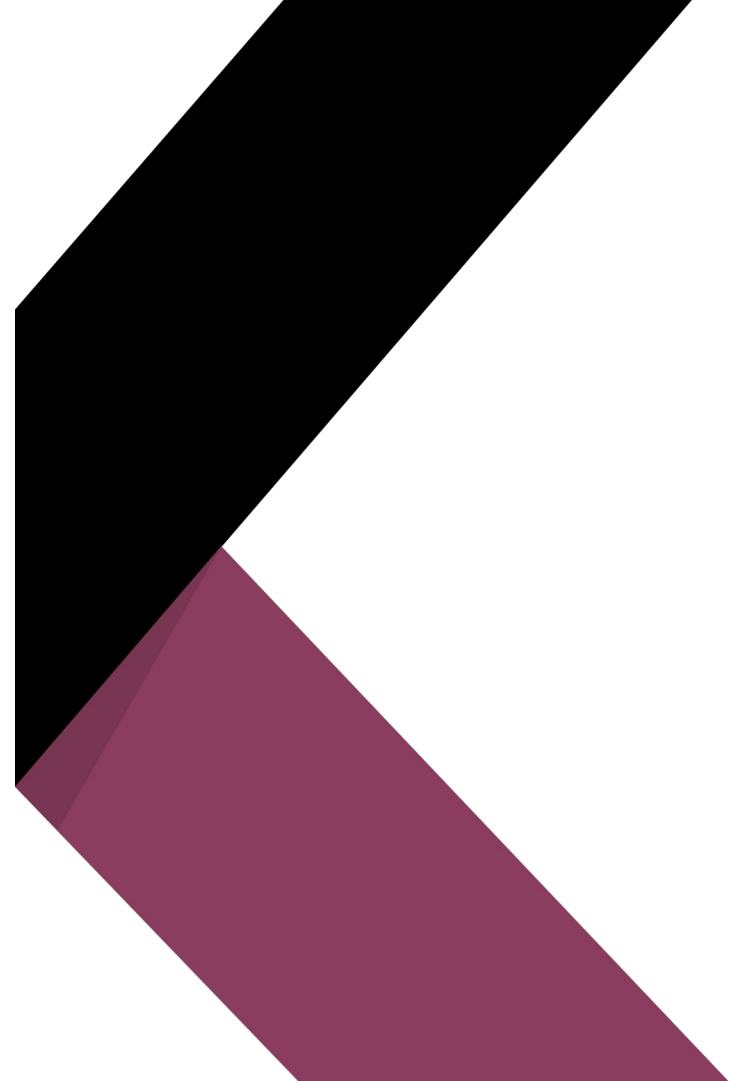


Battery
Electric Vehicles

BEV Induction



Battery Electric Vehicle

It's a *Pure Electric Vehicle* where the only traction device is the Electric Motor.

The Electric Motor only uses chemical energy supplied by Rechargeable Battery Packs.

BEV main components are:

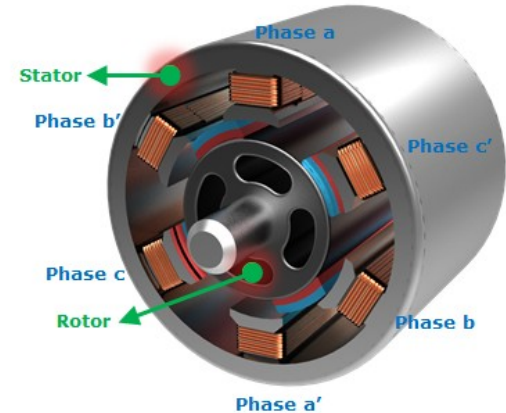
1. Electric Machine
2. Battery Packs
3. Power Managing Units
4. Digital Control System



Electric Motor

Electric Machine

- Since the motor is an electric one, there is NO transmission.
- When the electric motor starts moving, it provides all the drive torque (it can reach engine speeds > 10000 rpm) with a proper velocity, **without** the need for any gear ratio.
- This is the reason why it doesn't need reverse gear as well! It is enough to reverse the motor current phase to make it operating in the opposite direction.
- So, the only two commands to drive it:
 1. Gas Pedal
 2. Brake Pedal
-





Regenerative Braking

Electric Machine

In order to save energy and make the HV battery charge as long-lasting as possible, it is *essential* to take advantage of the **bi-directional operation** of the electric motor.

When the driver presses the brake pedal, the discs and the brake pads create friction as they meet. In turn, the friction creates kinetic energy that dissipates into the environment in the form of heat.

Regenerative braking recovers some of the kinetic energy that would otherwise turn into heat; the electric motor reverses its operation, starts working as an alternator and converts the kinetic energy into electric current;

There are two conditions for regeneration occurrence:

1. The driver applies the brake pedal → **Regenerative Braking**
2. The driver releases the accelerator pedal, and the vehicle is coasting → **E-Pedal** Functionality or **One-Pedal-Drive**

The amount of electricity the system generates is proportional to the level of braking force; the stronger the braking force, the greater the electrical current.

Ultimately, the amount of energy the system captures is dependent on the vehicle's speed and the duration of brake application.



E-Pedal Functionality

Electric Machine

The E-Pedal (or *One Pedal Drive*) functionality it's a extension of the Regenerative Braking concept broadened to the gas pedal release.

When the driver releases the accelerator pedal, the electric motor reverses its operation and starts working as an alternator.

It converts the rotational kinetic energy at wheel level into electric current;

Electric current to recharge the HV Battery Pack.

This means energy dissipation at torque level; practically, the vehicle slows down until it stops.

This is the reason why it is also called **ONE PEDAL DRIVE**; when the *E-Pedal* functionality is on, the driver doesn't need the brake pedal to stop the vehicle.

It's enough to release the accelerator pedal and, depending on the current vehicle speed and the slackening on the gas pedal, the vehicle coast down will be modulate, stopping the vehicle and giving motion to the alternator by mechanical energy.

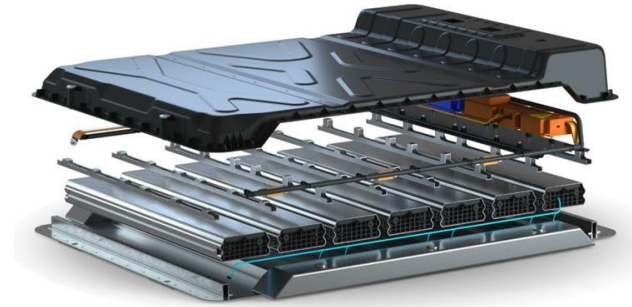




Battery System

Battery Packs

- An electric vehicle battery (EVB, also known as a traction battery) is an **High Voltage rechargeable battery** used to power the electric motors. The most common battery type in modern electric vehicles are **lithium-ion batteries** because of their high energy density compared to their weight. All-electric cars have battery capacities between 6.0 kWh (2012 Renault Twizy) and 100 kWh (2012 Tesla Model S and 2015 Tesla Model X).
- However, every electric vehicle has another **Low Voltage auxiliar battery**; Usually, it is a 12V battery used to power most of vehicle actuators and all the vehicle control units. It takes its charge from the HV Battery which is the only one directly connected to the grid (passing by the Onboard Charger)





Click
to
ad
d
Te
xt



Power Electronics

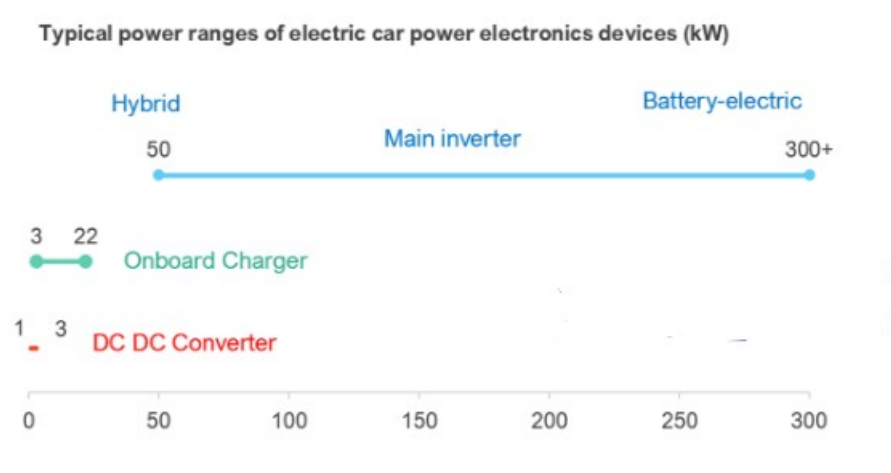
Power Managing Units

BEV Power Electronics include all the components between the Electric Motor and the Battery Pack used to regulate (as a filter) the energy transmission based on the instantaneous energy demand.

Main components are:

- **Inverter:** a high-power DC to AC converter for the battery to power the traction motor
- **Onboard Charger:** an AC - DC rectifier to charge the battery from the electric grid
- **DC-DC Converter:** for the high-voltage traction battery to power a low-voltage battery (typically, a 12V Battery for hotel facilities and ECU systems)

Most critical of all is the main inverter, which operates at the highest power and facilitates traction.



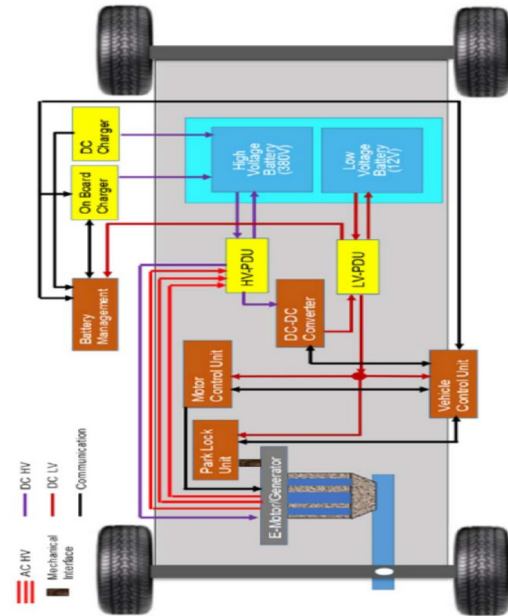


Main Control Units

Digital Control System

Every Battery Electric Vehicle has some really typical control units:

- **PIM** (Power Inverter Module): high-power DC to AC converter for the battery to power the traction motor. It includes:
 1. **EVCU** (Electric Vehicle Control Unit): to manage Propulsion, Devices Temperatures, Auxiliary Components, Battery Charge, and main vehicle actuators.
 2. **MCP** (Motor Control Processor): to manage; manage E-Motor kinematic parameters. Starting from gas & brake pedal press informations, MCP strictly interacts with EVCU to report the estimated shaft torque for torque compensation.
- **IDCM** (Integrated Dual Charger Module): it combines an on-board charger (AC\DC Inverter to charge the vehicle from the electric grid) and DC/DC converter (to convert the HV electric current in a 12V Electric Current to power the Auxiliary 12V Battery and all the Control Units) in one
- **BPCM** (Battery Pack Control Module): it is responsible for the management of critical features such as Voltage, Temperature and Current monitoring, Battery state of charge (SoC) and cell balancing of lithium-ion (Li-ion) batteries.





332 City BEV **VS** **595e Abarth BEV**

Fiat 500 – 332 City BEV is the first fully electric car from Fiat Chrysler Automobiles (FCA)., unveiled in March 2020.

Its Sporty version, **595e Abarth BEV**, will be release in March 2023 and its going to be faster, quicker, using the same electric motor but different regulation, firmer suspensions and more advanced stability control systems.





332 Drive Styles

332 City BEV

As the name suggests, 332 City BEV was meant to be city car, funny but not sporty.

This is the reason why, the driver can choose between the following 3 drive styles using a selector device:

1. **NORMAL** drive style: to drive everywhere with good accelerations and a 150 kph maximum speed
2. **RANGE** drive style: for city driving, with slightly slower acceleration and the One-Pedal-Drive functionality active
3. **SHERPA** drive style: to preserve the vehicle Battery SOC, with very slow acceleration, One-Pedal-Drive active and 80 kph as maximum

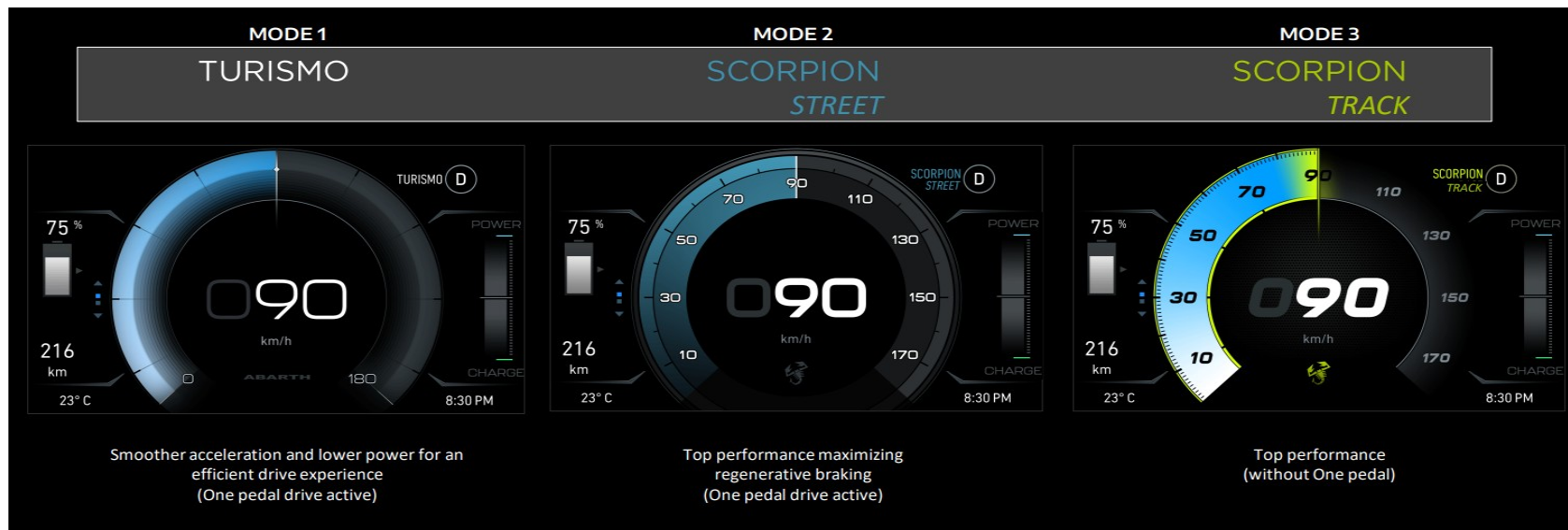




595e New Drive Styles

595e Abarth BEV

The 595e Abarth BEV concept will be different, with only one urban meant drive style , called Turismo on IPC, and two other Aggressive Drive Styles:





Different Features

332 vs 595e

Features	332 City BEV	595e Abarth BEV
Lithium Ion Battery Pack	42 kWh → 320 km by WLTP (400km)	42 kWh → 225 km by WLTP cycle
	24 kWh → 180km by WLTP	
Front Axle Permanent Magnet Synchronous Motor	$P_{\max} = 87 \text{ kW (118 CV)}$	$P_{\max} = 120 \text{ kW (180 CV)}$
Torque	$T_{\max} = 2112 \text{ Nm}$	$T_{\max} = 2400 \text{ Nm}$
	$\tau = 9,8$	$\tau = 10,22$
Vehicle Size	$r_{\max} = 28,8 \text{ cm}$	$r_{\max} = 29,2 \text{ cm}$
	$m = 1433 - 1493 \text{ kg}$	$m = 1464 - 1524 \text{ kg}$



Driver Selectable Drive Styles

332 vs 595e

CAN C1 BODY7.DriveStyleSts	332 City BEV	595e Abarth BEV
Normal (0)	Normal OPD Not Active	Sport Street OPD Active
City (1)	Range OPD Active	Turismo OPD Active
Sport Fun (2)	-	Sport Track OPD Not Active
Eco (4)	Sherpa OPD Active	-



Driver Independent Drive Modes

332 vs 595e

SOC-related Drive Mode	332 City BEV	595e Abarth BEV
Turtle1 <i>16km < EstRange < 24km</i>	<i>Previous Drive Style Speed Limitation</i>	<i>Previous DriveStyle Speed Limitation</i>
Turtle2 <i>8km < EstRange < 16km</i>	<i>Previous Drive Style Speed Limitation</i>	<i>Previous DriveStyle Speed Limitation</i>
Turtle3 <i>0km < EstRange < 8km</i>	$v_{max} = 70 \text{ kph}$	$v_{max} = 70 \text{ kph}$
Turtle4 <i>EstRange <= 0km</i>	$v_{max} = 70 \text{ kph}$	$v_{max} = 70 \text{ kph}$



Driver Independent Drive Modes

332 vs 595e

OPD-Disabling Drive Mode	<i>332 City BEV</i>	<i>595e Abarth BEV</i>
Non-Aggressive Pedal Map w/o One-Pedal-Drive	Normal Pedal Map	Default Pedal Map <i>(P.N. in HCP sw Default Pedal Map has been calibrated on the ex-Eco Pedal Map in both DTRR and STMR rings, since 595 doesn't provide this DriveStyle)</i>

CONTACTS

Via G. Porzio,
Centro Direzionale Isola B3
80143 Napoli Via G. Porzio

www.kineton.it

